Original Article

Incidence of postoperative pain after single-visit and multiple-visit root canal therapy: A randomized controlled trial

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INTRODUCTION

Root canal therapy (RCT) requires removing organic tissue, debris, and pathogenic microbes from the root canal system utilizing mechanical instrumentation associated with copious irrigation with disinfectant agents. Two approaches are advocated to unravel this problem. In the first approach, remaining bacteria are eradicated or prohibited from repopulating the root canal system by giving an inter-appointment dressing into the root canal. However, even a negative culture prior to obturation does not guarantee healing in all situations.[1]

Another approach aims to eliminate residual microbes or render them unhazardous by entombing. Complete and three-dimensional obturation deprives the microbes of nourishment and the space required to thrive and reproduce. The treatment is completed in one visit. The sealer's antimicrobial activity or the zinc ions of gutta-percha can destroy the remaining bacteria.[1]

Short-term complications after root canal treatment include postoperative inflammation of periapical tissues leading to mild pain or flare-up. Patients might regard

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postoperative pain and flare-up as a yardstick by which the operator's abilities are assessed. The pain perception depends on the tissue damage severity, and the persistence of the injury source determines the outcome of RCT.[2]

The evidence for recommending either one or multiple visits (MV) to RCT is not consistent. Moayad Ahmed Alomaym *et al*.[3] assessed the incidence and severity of postobturation pain and deduced that there was a statistically significant less incidence of pain in MV groups than single–visit (SV) one. Singh *et al*.[4] inferred that the mean pain score in the SV was lower than that of the MV group. However, the difference was statistically insignificant. Other studies[5,6] concluded that there was no significant difference in postoperative pain between SV and MV treatment.

As the usage of newer rotary nickel–titanium instruments increases, a randomized clinical trial comparing SV and MV RCT, both done using such instruments, is desirable. Hence, this study primarily aims to compare postobturation pain prevalence at 6-, 12-h, 1-, and 2-day after SV versus two-visit RCT on the mandibular first molar. The secondary aim is to compare tenderness on percussion after 1 week of treatment.

MATERIALS AND METHODS

Institutional Ethics Committee clearance was acquired, and the study was registered with the Clinical Trials Registry - India (CTRI/2019/05/019067).

Selection of subjects

Ninety patients seeking RCT of mandibular first molars were screened. Fifteen individuals did not meet the criteria for inclusion, and five patients did not consent to participate. Seventy patients were recruited, and their informed consent was obtained.

Inclusion criteria

- The patient should consent to the planned SV or two visit treatment
- Moderately curved root canals.

Exclusion criteria

- Patients having any systemic disease
- Pregnant patients
- Teeth with calcified canals, unusual canal morphology, teeth with extra root or procedural error during treatment
- Teeth with periapical radiolucency of size >0.5 cm
- A patient who took antibiotics and/or analgesics right after the first appointment of the therapy.

Sample size

The sample size was estimated using the formula: $n = \frac{(Z\alpha + Z\beta)^2 2s^2}{(\bar{x}1 - \bar{x}2)^2}$ taking the significance level at 5% and the test's power at 80%. The sample size estimated was 30 patients per group and was appraised as 35 patients per group to compensate for dropouts during follow-up.

Randomization

A parallel group trial design was used for assigning the patients to the two study groups depending on number of treatment appointments. A random list was generated, 35 patients received treatment in SV (Group 1) and 35 patients in two-visit (Group 2).

Intervention

The principal investigator completed RCT in all cases.

The investigator explained the treatment procedures. Then administered inferior alveolar nerve block using 2% lidocaine with 1:80,000 epinephrine to anesthetize the tooth. After isolation of tooth using a rubber dam, the operator performed access preparation. Canals were negotiated, and apical patency was preserved with a number 10 K-file. The working length was determined with K-file using an apex locator (Root ZX, J Morita corp.) and confirmed by a periapical radiograph. Cleaning and shaping were performed with a hybrid technique using hand K-files (Mani Inc., Japan) and S-one NiTi rotary files (Fanta) for all the teeth. Instrumentation was carried out using 0.04 and 0.06 taper NiTi rotary files along with copious irrigation using 3% NaOCl and saline. Files were employed at a speed of 350 rpm with a slow, gentle in and out movement. 3% Sodium hypochlorite and 17% EDTA were utilized during canal preparation and saline as the final irrigant. Files at least three sizes larger than the initial apical file were used for apical preparation. The canals were dried utilizing sterile absorbent points.

Teeth in Group 1 were obturated at the same visit with gutta-percha cones and Sealapex sealer utilizing the lateral compaction technique, and temporary restoration was placed. A postobturation radiograph was taken. Ibuprofen was prescribed as a rescue medication, and the patients were advised to take it only if they had severe pain after treatment.

In Group 2, calcium hydroxide dressing (RC Cal, Prime Dental Products Pvt. Ltd., India) was placed in the canal as an interappointment medicament. Teeth were sealed with a sterile cotton pellet and a provisional filling material (Prime TMP-RS eugenol free). After 1 week, teeth were obturated utilizing a similar method and materials as used for Group 1. Ibuprofen was prescribed as a rescue medication, and the patients were told to take it only if they had severe pain after treatment.

If a patient in either group took rescue medication within 48 hours after the treatment, he/she was excluded from the study. After 1 week of obturation, teeth were restored by composite restoration (Herculite précis, Kerr, Asia).

Pain assessment

Patients recorded their preoperative pain levels using heft-parker visual analog scale (HP VAS) in the clinician's presence to make sure that they understood the instructions. The patient carried the HP VAS form along with them to record postoperative pain levels at 6-, 12-h, 1, and 2 days' intervals. A telephonic reminder was given to them to note their pain readings and return the form duly filled. The treated teeth were evaluated for tenderness to percussion after 1 week of obturation.

Statistical analysis

the arch's confounding effect. Tooth location may affect

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